WoSM Academy

CS102-F26 · DIGITAL DESIGN AND COMPUTER ARCHITECTURE I

Timing of Sequential Circuits

Lecture 14: setup time, hold time and how fast the clock can go

Dr. Ali Khudiyev

Fall 2026

CC BY-SA 4.0

Today

01 Trouble at the edge

02 Setup time constraint

03 Hold time constraint

04 Clock skew

05 Summary

01

Trouble at the edge

What if D changes at the edge?

- In the master–slave flip-flop (Unit 12), the rising edge closes the master and opens the slave. But what if D changes at the edge?

D changes just after the edge

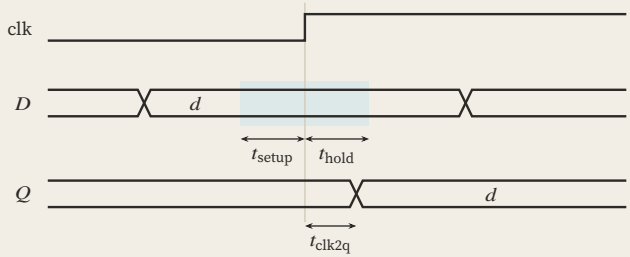
The master is enabled by clk′, which comes through a NOT gate. For a moment after the edge, clk′ is still 1 while clk is already 1: both latches are transparent, and the new D runs straight through to Q.

D changes just before the edge

The master closes while its gates are still settling on the new D. Q_m can end up 0 or 1, or swing between them for a while: the flip-flop is metastable, as the SR latch was in Lecture 11.

- The book's simulations (Figures 5.11 and 5.12) show both: Q takes the wrong value, or oscillates.
- The cure: keep D steady for a short while around every rising edge.

Setup time and hold time



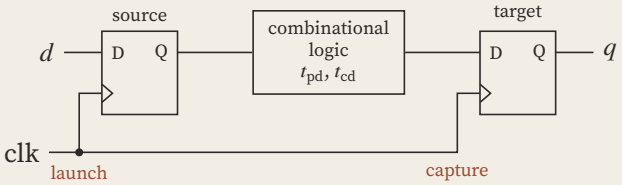
Definition (Timing of a flip-flop)

Setup time t_{setup}: how long D must be stable *before* the clock edge. *Hold time* t_{hold}: how long it must stay stable *after* it. *Clock-to-Q delay* t_{clk2q}: how long after the edge the new value appears at Q.

02

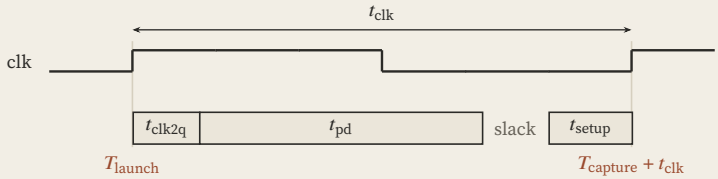
Setup time constraint

From one flip-flop to the next



- The basic pattern of every synchronous circuit: a value leaves the *source* flip-flop at one clock edge, passes through combinational logic, and is stored by the *target* flip-flop at the next edge.
- t_{pd} and t_{cd} are the logic's propagation and contamination delays `(Unit 9)`: its longest and its shortest path.
- The edge reaches the source at time T_{launch} and the target at $T_{capture}$.

The setup time constraint



The new value must be at the target's D at least t_{setup} before the *next* edge, even along the longest path:

$$T_{launch} + t_{clk2q} + t_{pd} + t_{setup} \;\leq\; T_{capture} + t_{clk}.$$

When the clock reaches both flip-flops at the same time ($T_{launch} = T_{capture}$):

$$t_{clk} \;\geq\; t_{clk2q} + t_{pd} + t_{setup}.$$

How fast can the clock go?

Example

$t_{\text{clk2q}} = 30$ ps, $t_{\text{pd}} = 420$ ps, $t_{\text{setup}} = 50$ ps.

$t_{\text{clk}} \geq 30 + 420 + 50 = 500$ ps, so the clock can run at up to $1/500$ ps $= 2$ GHz.

- The slowest path between any two flip-flops sets the clock period of the whole circuit.
- Too slow? Run the clock slower, make the logic on that path faster, or put another register in the middle of it: the idea behind pipelining (Unit 20).

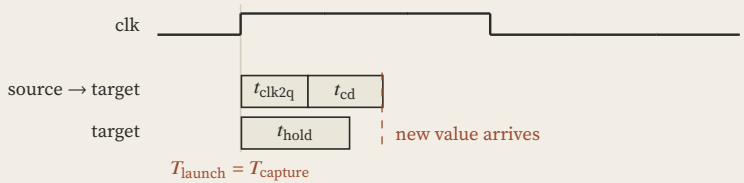
Remark

The synchronous counter of Lecture 13: the longest path runs from y_0 through the chain of AND gates to the last T input. The more bits, the longer the chain, the slower the clock.

03

Hold time constraint

The hold time constraint



The same edge that the target is still capturing also launches the next value from the source. Along the *shortest* path, it must not arrive before the target's hold time is over:

$$T_{launch} + t_{clk2q} + t_{cd} \;\geq\; T_{capture} + t_{hold}, \qquad \text{or, with no skew,} \qquad t_{clk2q} + t_{cd} \;\geq\; t_{hold}.$$

Fixing a hold violation

- t_{clk} does not appear in the hold constraint. A circuit that violates it fails at every clock speed: slowing the clock does not help.
- The fix is to make the shortest path slower: add buffers (or other gates that keep the value) on it.
- Two flip-flops wired directly, with no logic between them, have $t_{\text{cd}} = 0$: they need $t_{\text{clk2q}} \geq t_{\text{hold}}$.

Example

$t_{\text{clk2q}} = 30$ ps, $t_{\text{cd}} = 20$ ps, $t_{\text{hold}} = 60$ ps. Then $30 + 20 = 50 < 60$: the hold time is violated by 10 ps. Adding at least 10 ps of delay to the shortest path fixes it, as long as the longest path still meets the setup constraint.

04

Clock skew

When the clock arrives late

- The clock is a signal on a wire too: it can reach the target later than the source. The difference is the *clock skew*, $t_{\text{skew}} = T_{\text{capture}} - T_{\text{launch}}$.
- Put it into the two constraints of the book:

$$\begin{aligned} \text{setup:} \quad & t_{\text{clk}} \;\geq\; t_{\text{clk2q}} + t_{\text{pd}} + t_{\text{setup}} - t_{\text{skew}} \\ \text{hold:} \quad & t_{\text{clk2q}} + t_{\text{cd}} \;\geq\; t_{\text{hold}} + t_{\text{skew}} \end{aligned}$$

- A late capture edge gives the longest path more time, but the shortest path must now wait even longer. Skew helps one constraint and hurts the other.

> **Remark**
>
> The book builds the DX8 without doing these checks, and its epilogue leaves them to the reader. You now know how.

Try it

1. $t_{\text{clk2q}} = 40$ ps, $t_{\text{pd}} = 160$ ps, $t_{\text{setup}} = 50$ ps. What is the fastest clock?
2. The same circuit has $t_{\text{cd}} = 15$ ps and $t_{\text{hold}} = 70$ ps. Does it work at 1 GHz?
3. Now the clock reaches the target 20 ps after the source. Redo both checks.
4. Two flip-flops wired directly, $t_{\text{clk2q}} = 30$ ps, $t_{\text{hold}} = 20$ ps. Is that safe? And with 15 ps of skew?

Try it

1. $t_{\text{clk2q}} = 40$ ps, $t_{\text{pd}} = 160$ ps, $t_{\text{setup}} = 50$ ps. What is the fastest clock?
   $t_{\text{clk}} \geq 250$ ps: 4 GHz.
2. The same circuit has $t_{\text{cd}} = 15$ ps and $t_{\text{hold}} = 70$ ps. Does it work at 1 GHz?

3. Now the clock reaches the target 20 ps after the source. Redo both checks.

4. Two flip-flops wired directly, $t_{\text{clk2q}} = 30$ ps, $t_{\text{hold}} = 20$ ps. Is that safe? And with 15 ps of skew?

Try it

1. $t_{\text{clk2q}} = 40$ ps, $t_{\text{pd}} = 160$ ps, $t_{\text{setup}} = 50$ ps. What is the fastest clock?
   $t_{\text{clk}} \geq 250$ ps: 4 GHz.
2. The same circuit has $t_{\text{cd}} = 15$ ps and $t_{\text{hold}} = 70$ ps. Does it work at 1 GHz?
   No: $40 + 15 = 55 < 70$. It fails at any clock speed; add at least 15 ps to the shortest path.
3. Now the clock reaches the target 20 ps after the source. Redo both checks.

4. Two flip-flops wired directly, $t_{\text{clk2q}} = 30$ ps, $t_{\text{hold}} = 20$ ps. Is that safe? And with 15 ps of skew?

Try it

1. $t_{\text{clk2q}} = 40$ ps, $t_{\text{pd}} = 160$ ps, $t_{\text{setup}} = 50$ ps. What is the fastest clock?
   $t_{\text{clk}} \geq 250$ ps: 4 GHz.
2. The same circuit has $t_{\text{cd}} = 15$ ps and $t_{\text{hold}} = 70$ ps. Does it work at 1 GHz?
   No: $40 + 15 = 55 < 70$. It fails at any clock speed; add at least 15 ps to the shortest path.
3. Now the clock reaches the target 20 ps after the source. Redo both checks.
   Setup: $t_{\text{clk}} \geq 230$ ps (about 4.35 GHz). Hold: $55 \geq 70 + 20$ fails by 35 ps.
4. Two flip-flops wired directly, $t_{\text{clk2q}} = 30$ ps, $t_{\text{hold}} = 20$ ps. Is that safe? And with 15 ps of skew?

Try it

1. $t_{\text{clk2q}} = 40$ ps, $t_{\text{pd}} = 160$ ps, $t_{\text{setup}} = 50$ ps. What is the fastest clock?
   $t_{\text{clk}} \geq 250$ ps: 4 GHz.
2. The same circuit has $t_{\text{cd}} = 15$ ps and $t_{\text{hold}} = 70$ ps. Does it work at 1 GHz?
   No: $40 + 15 = 55 < 70$. It fails at any clock speed; add at least 15 ps to the shortest path.
3. Now the clock reaches the target 20 ps after the source. Redo both checks.
   Setup: $t_{\text{clk}} \geq 230$ ps (about 4.35 GHz). Hold: $55 \geq 70 + 20$ fails by 35 ps.
4. Two flip-flops wired directly, $t_{\text{clk2q}} = 30$ ps, $t_{\text{hold}} = 20$ ps. Is that safe? And with 15 ps of skew?
   Without skew, yes: $30 \geq 20$. With it, no: $30 < 20 + 15$.

05

Summary

Summary

- A flip-flop needs D stable from t_{setup} before the clock edge until t_{hold} after it; otherwise it can store the wrong value or go metastable. Its new value appears t_{clk2q} after the edge.
- Setup: the longest path must fit in one period, $t_{\text{clk}} \geq t_{\text{clk2q}} + t_{\text{pd}} + t_{\text{setup}}$. This sets the fastest clock.
- Hold: the shortest path must not be too quick, $t_{\text{clk2q}} + t_{\text{cd}} \geq t_{\text{hold}}$, at any clock speed.
- Clock skew loosens the setup constraint and tightens the hold constraint.

Before the next lecture

- Read Section 5.5 of the book.
- For the 3-bit synchronous counter of Lecture 13, list every path from a flip-flop output to a flip-flop input. Which one is the longest, and which the shortest?
- Read the beginning of Chapter 6 and Section 6.1.

> **Next lecture** (Unit 15)
>
> Instruction set architecture: the Von Neumann architecture, and the DX8's instruction format and instruction set.